Thresholding Operators for Complex Images

1 Problem and Reduction to a Magnitude Problem

Let $\mathbf{z} = (z_1, \dots, z_N)^\mathsf{T} \in \mathbb{C}^N$ be the observed complex image and $\mathbf{x} \in \mathbb{C}^N$ the estimate. For $w \in \mathbb{C}$, $|w|$ is the modulus and $\operatorname{sgn}(w) = w/|w|$ ($\operatorname{sgn}(0) = 0$). For $\lambda > 0$,

$$\hat{\mathbf{x}} = \operatorname*{arg\,min}_{\mathbf{x} \in \mathbb{C}^N} \tfrac{1}{2}\|\mathbf{x} - \mathbf{z}\|_2^2 + \lambda R(\mathbf{x}), \tag{1}$$

where $\|\mathbf{v}\|_2^2 = \sum_i |v_i|^2$ and $R : \mathbb{C}^N \to [0, \infty)$ is a regularizer. The weight λ is real: the objective must be real-valued to be minimized, since $\mathbb{C}$ has no ordering, and $\lambda > 0$ sets the trade-off between the data term and the nonnegative penalty.

Lemma 1 (Separability)**.** Assume R is separable over the entries, $R(\mathbf{x}) = \sum_{i=1}^N R_i(x_i)$ with $R_i : \mathbb{C} \to [0, \infty)$. The data term is separable as well, $\|\mathbf{x} - \mathbf{z}\|_2^2 = \sum_i |x_i - z_i|^2$, so the objective of (1) is $\sum_i J_i(x_i)$ with $J_i(x) = \frac{1}{2}|x - z_i|^2 + \lambda R_i(x)$, and each $\hat{x}_i = \arg\min_{x \in \mathbb{C}} J_i(x)$ is found independently.

Drop the index i.

Lemma 2 (Phase)**.** Assume further that each R_i depends only on the modulus, $R_i(x) = r(|x|)$ for some $r : [0, \infty) \to [0, \infty)$ with $r(0) = 0$. This holds for the penalties used here: a norm on $\mathbb{C}^N$ is absolutely homogeneous, $\|\alpha\mathbf{x}\| = |\alpha|\,\|\mathbf{x}\|$ for $\alpha \in \mathbb{C}$, so with $\alpha = e^{j\theta}$ it is invariant to phase rotation; $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$ and $\|\mathbf{x}\|_{1/2}^{1/2}$ are all defined through $|x_i|$. Then, for a fixed modulus $|x|$, the penalty is constant and $|x - z|$ is smallest when x has the phase of z, where

$$|x - z|^2 = (|x| - |z|)^2. \tag{2}$$

Proof in polar form. Write $x = \rho e^{j\theta}$ and $z = m e^{j\varphi}$, with $\rho, m \geq 0$. By the law of cosines,

$$|x - z|^2 = \rho^2 + m^2 - 2\rho m \cos(\theta - \varphi) \;\geq\; (\rho - m)^2, \tag{3}$$

with equality iff $\cos(\theta - \varphi) = 1$, i.e. $\theta = \varphi$ (for $\rho, m > 0$). □

Proof in rectangular form. Write $x = a + jb$ and $z = c + jd$. Then

$$|x - z|^2 = |x|^2 + |z|^2 - 2\operatorname{Re}(x\bar{z}), \qquad \operatorname{Re}(x\bar{z}) = ac + bd.$$

By the Cauchy–Schwarz inequality, $ac + bd \leq \sqrt{a^2 + b^2}\,\sqrt{c^2 + d^2} = |x|\,|z|$, so $|x - z|^2 \geq (|x| - |z|)^2$, with equality iff $(a, b) = \kappa\,(c, d)$ for some $\kappa \geq 0$, i.e. x is a nonnegative multiple of z and has its phase. □

Combining Lemmas 1 and 2, the minimizer keeps the phase of z and thresholds its magnitude:

$$\hat{x} = \operatorname{sgn}(z)\,\hat{\rho}, \qquad \hat{\rho} = \operatorname*{arg\,min}_{\rho \geq 0}\; g(\rho), \qquad g(\rho) = \tfrac{1}{2}(\rho - m)^2 + \lambda r(\rho), \quad \rho = |x|,\; m = |z|. \tag{4}$$

2 ℓ_1: Soft Thresholding

$R(\mathbf{x}) = \|\mathbf{x}\|_1 = \sum_i |x_i|$ satisfies Lemmas 1 and 2 with $r(\rho) = \rho$. Then $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\rho$ is convex on $\rho \geq 0$, with $g'(\rho) = \rho - m + \lambda$. If $m > \lambda$, then $g' = 0$ at $\rho = m - \lambda > 0$. If $m \leq \lambda$, then $g' > 0$ for all $\rho > 0$, so the minimizer is $\rho = 0$. Hence $\hat{\rho} = \max(m - \lambda, 0)$:

$$\mathcal{S}_\lambda(z) = \operatorname{sgn}(z)\,\max(|z| - \lambda,\, 0) = \begin{cases} (|z| - \lambda)\,\dfrac{z}{|z|}, & |z| > \lambda, \\ 0, & |z| \leq \lambda. \end{cases} \tag{5}$$

Retained entries are shrunk by λ in magnitude.

3 ℓ_0: Hard Thresholding

$R(\mathbf{x}) = \|\mathbf{x}\|_0 = \#\{i : x_i \neq 0\}$ counts the nonzero entries. Since $x_i \neq 0 \Leftrightarrow |x_i| > 0$, it satisfies Lemmas 1 and 2 with $r(\rho) = \mathbf{1}[\rho > 0]$, and $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\,\mathbf{1}[\rho > 0]$. Compare the two cases: $\rho = 0$ gives $g(0) = \frac{1}{2}m^2$; for $\rho > 0$ the penalty is constant, so the best value is $\rho = m$, with $g(m) = \lambda$. $\rho = m$ is better iff $\lambda < \frac{1}{2}m^2$, i.e. $m > \sqrt{2\lambda}$:

$$\mathcal{H}_{\sqrt{2\lambda}}(z) = \begin{cases} z, & |z| > \sqrt{2\lambda}, \\ 0, & |z| \leq \sqrt{2\lambda}. \end{cases} \tag{6}$$

At $|z| = \sqrt{2\lambda}$, both 0 and z are minimizers; (6) picks 0. Retained entries are unchanged.

4 $\ell_{1/2}$: Half Thresholding

$R(\mathbf{x}) = \|\mathbf{x}\|_{1/2}^{1/2} = \sum_i |x_i|^{1/2}$ satisfies Lemmas 1 and 2 with $r(\rho) = \sqrt{\rho}$, and $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\sqrt{\rho}$. Let $\mu = 2\lambda$ and minimize $2g(\rho) = (\rho - m)^2 + \mu\sqrt{\rho}$.

Stationary Point. For $\rho > 0$, $\frac{d}{d\rho}\big[(\rho - m)^2 + \mu\sqrt{\rho}\big] = 2(\rho - m) + \frac{\mu}{2\sqrt{\rho}}$. With $u = \sqrt{\rho}$, setting it to zero gives

$$u^3 - m\,u + \tfrac{\mu}{4} = 0. \tag{7}$$

Its largest root, the local minimum of g, is $u = 2\sqrt{m/3}\,\cos(\psi/3)$ with $\psi = \arccos\big(-\frac{\mu}{8}(\frac{m}{3})^{-3/2}\big)$. With $\phi = \pi - \psi$,

$$\rho^* = u^2 = \frac{2m}{3}\Big[1 + \cos\Big(\frac{2\pi}{3} - \frac{2}{3}\phi\Big)\Big], \qquad \phi = \arccos\Big(\frac{\mu}{8}\Big(\frac{m}{3}\Big)^{-3/2}\Big). \tag{8}$$

Cut-Off. ρ^* beats $\rho = 0$ iff $(\rho^* - m)^2 + \mu\sqrt{\rho^*} < m^2$. At the cut-off $m = t$, equality holds together with (7). Substituting $m = u^2 + \mu/(4u)$ from (7):

$$\big(\tfrac{\mu}{4u}\big)^2 + \mu u = \big(u^2 + \tfrac{\mu}{4u}\big)^2 \iff u^3 = \tfrac{\mu}{2},$$

so

$$t = u^2 + \frac{\mu}{4u} = \frac{3}{2}\Big(\frac{\mu}{2}\Big)^{2/3} = \frac{\sqrt[3]{54}}{4}\,(2\lambda)^{2/3}, \tag{9}$$

and at the cut-off $\rho^* = (\mu/2)^{2/3} = \frac{2}{3}t$.

Result.

$$\mathcal{T}_\lambda(z) = \begin{cases} \operatorname{sgn}(z)\,\dfrac{2|z|}{3}\Big[1 + \cos\Big(\dfrac{2\pi}{3} - \dfrac{2}{3}\phi(|z|)\Big)\Big], & |z| > t, \\ 0, & |z| \leq t, \end{cases} \qquad \phi(m) = \arccos\Big(\frac{\lambda}{4}\Big(\frac{m}{3}\Big)^{-3/2}\Big). \tag{10}$$

The output jumps from 0 to $\frac{2}{3}t$ at the cut-off, and the shrinkage $m - \rho^* \approx \lambda/(2\sqrt{m})$ vanishes for large m.

5 Summary

Each operator keeps the phase and thresholds the magnitude, $\hat{x} = \operatorname{sgn}(z)\,f(|z|)$:

	cut-off t on $\lvert z\rvert$	$\lvert\hat{x}\rvert$ for $\lvert z\rvert > t$	jump at t
ℓ_1	λ	$\lvert z\rvert - \lambda$	none
ℓ_0	$\sqrt{2\lambda}$	$\lvert z\rvert$	$0 \to t$
$\ell_{1/2}$	$\frac{\sqrt[3]{54}}{4}(2\lambda)^{2/3}$	Eq. (8)	$0 \to \frac{2}{3}t$